Computer networks

Lecture-3

Recap of Lecture 2

- Key Data Communication Terminology
- Data Flow/ Transmission Modes
- Networks and Why we need Them?
- Distributed Processing
- Network Criteria
- Physical structure of Network
  - Type of connection

Overview of Lecture 3

- Physical Structure of Network
  - Physical Topologies
    1. Bus
    2. Mesh
    3. Star
    4. Ring
    5. Tree
    6. Hybrid

Topology

"The Topology is the geometric representation of the relationship of the links and linking devices"

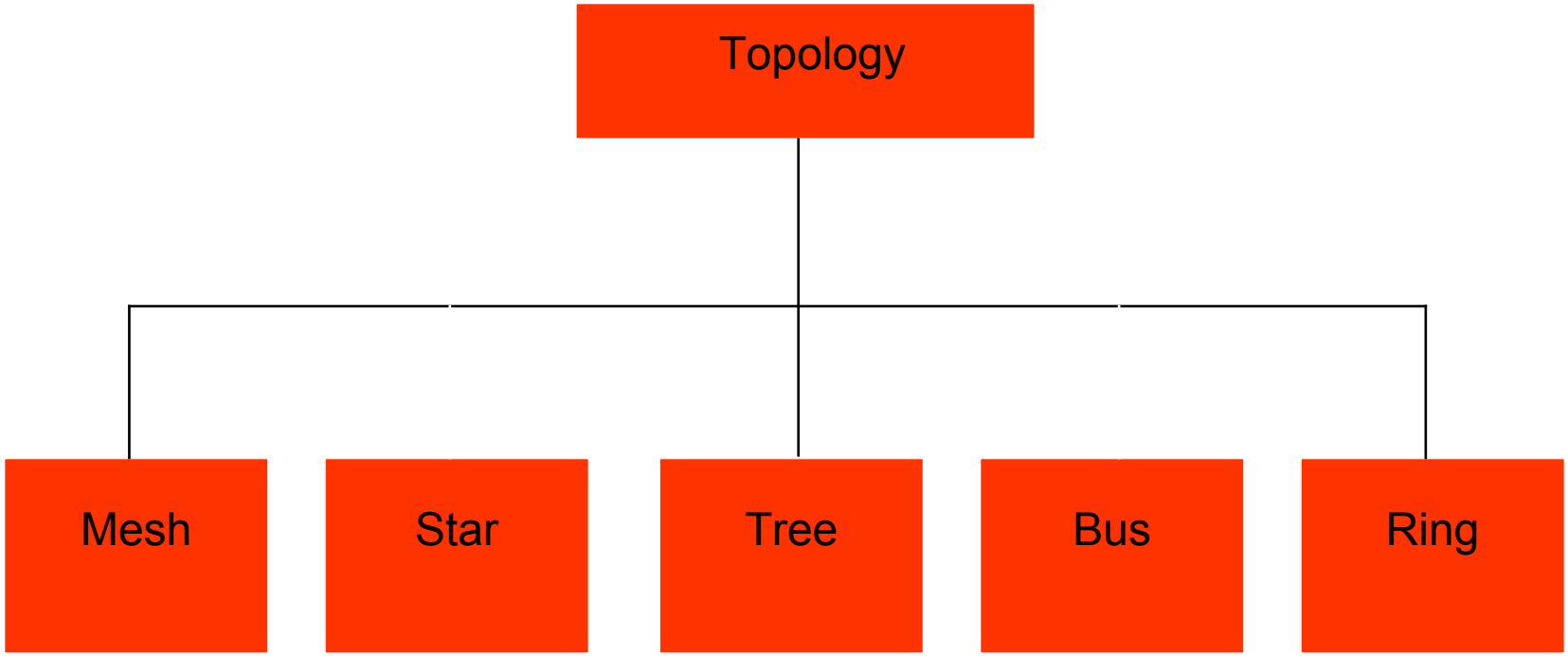
OR

"Topology defines physical or logical arrangement of links in a Network"

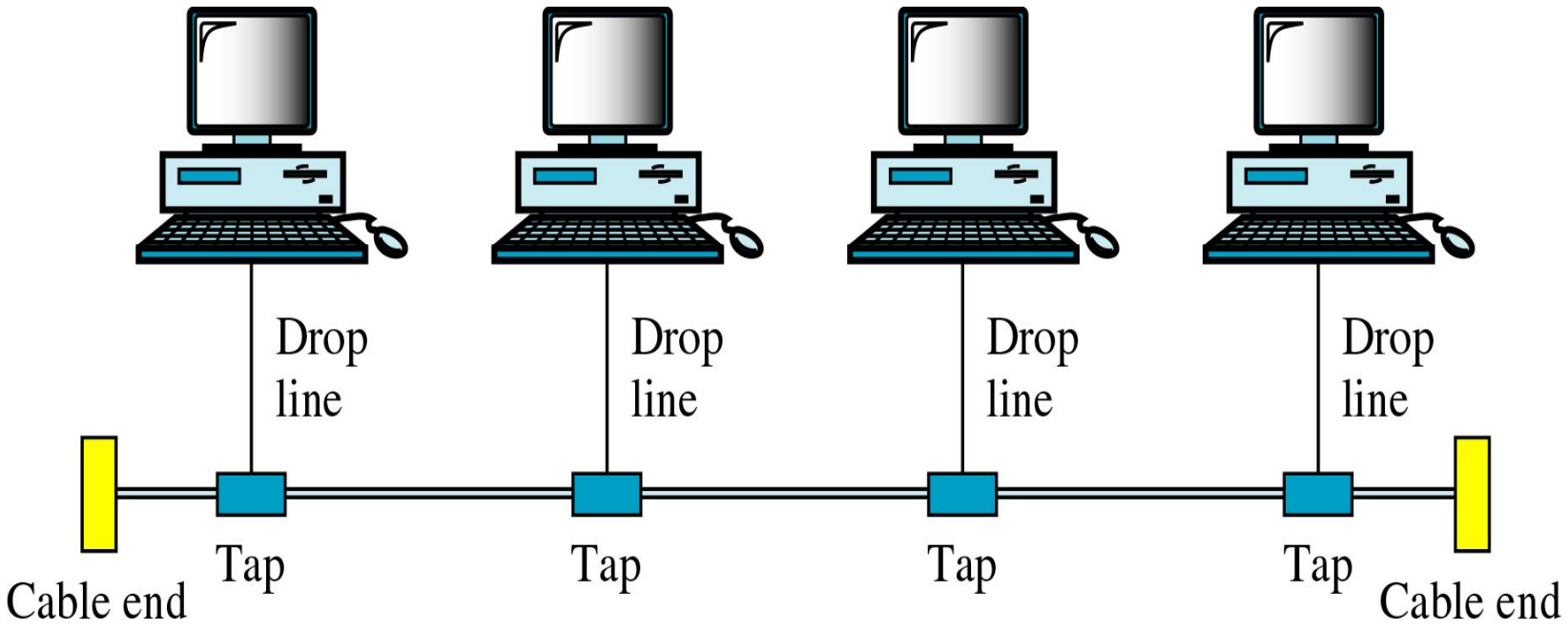
Topology (Cont.)

- The ways in which the wires can be run in the network to link with the computer.
- The ways in which the connections are made in the network.
- Simply The physical layout of the network.

Categories of Topology



Bus Topology



Bus Topology (Cont.)

- **Structure**
  - A bus topology, is multipoint.
  - One long cable acts as a backbone to link all the devices in a network.
  - Nodes are connected to the bus cable by drop lines and taps.
  - A drop line is a connection running between the device and the main cable.
  - A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
  - As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

Bus Topology (Cont.)

- In a bus network every workstation needs
  - BNC Barrel Connectors
  - T-Connectors
  - Terminators
  - NIC.
- **T-Connectors** are used to provide connection from the bus link to a workstation.
- **NIC** (Network Interface Card) is used to link a workstation with the network.
- **Barrel Connectors** are used to connect two parts of a cable either broken or when additional wire is required.
- **Terminators** are used for Termination purpose.
- Basically Thin Coaxial Cable is used in this topology.

Bus Topology (Cont.)

- **Uses:**
  - It is often used when network installation is small, simply or temporary.
  - Bus is a passive topology,
    - i.e. no active electron is used to amplify the signal.

Bus Topology (Cont.)

- **WORKING:**
  - Broadcast network
    - all the computers on the network receive the information, but only one computer (the one with the address that matches the one encoded in the message) accepts the information. The rest disregards the message.
  - Protocol used is ***CSMA/CD*** **(Carrier Sense Multiple Access with Collision Detection),**
    - i.e. only one device can transmit the data at a time. More than one signal introduces collision. It means if one computer sends a signal over the bus; the other computer must wait for some time.

Bus Topology (Cont.)

- Another important issue in bus network is ***termination***.
- Without termination, when the signal reaches the end of the wire, it bounces back and travels back up the wire which makes the network busy.
- To stop the bouncing signal, we attach terminators at either end of segment.
- The terminator absorbs the electrical energy and stops the reflection.
- **Example**
  - ***Ethernet 10Base2, 10Base5,*** and ***ARC Net***.

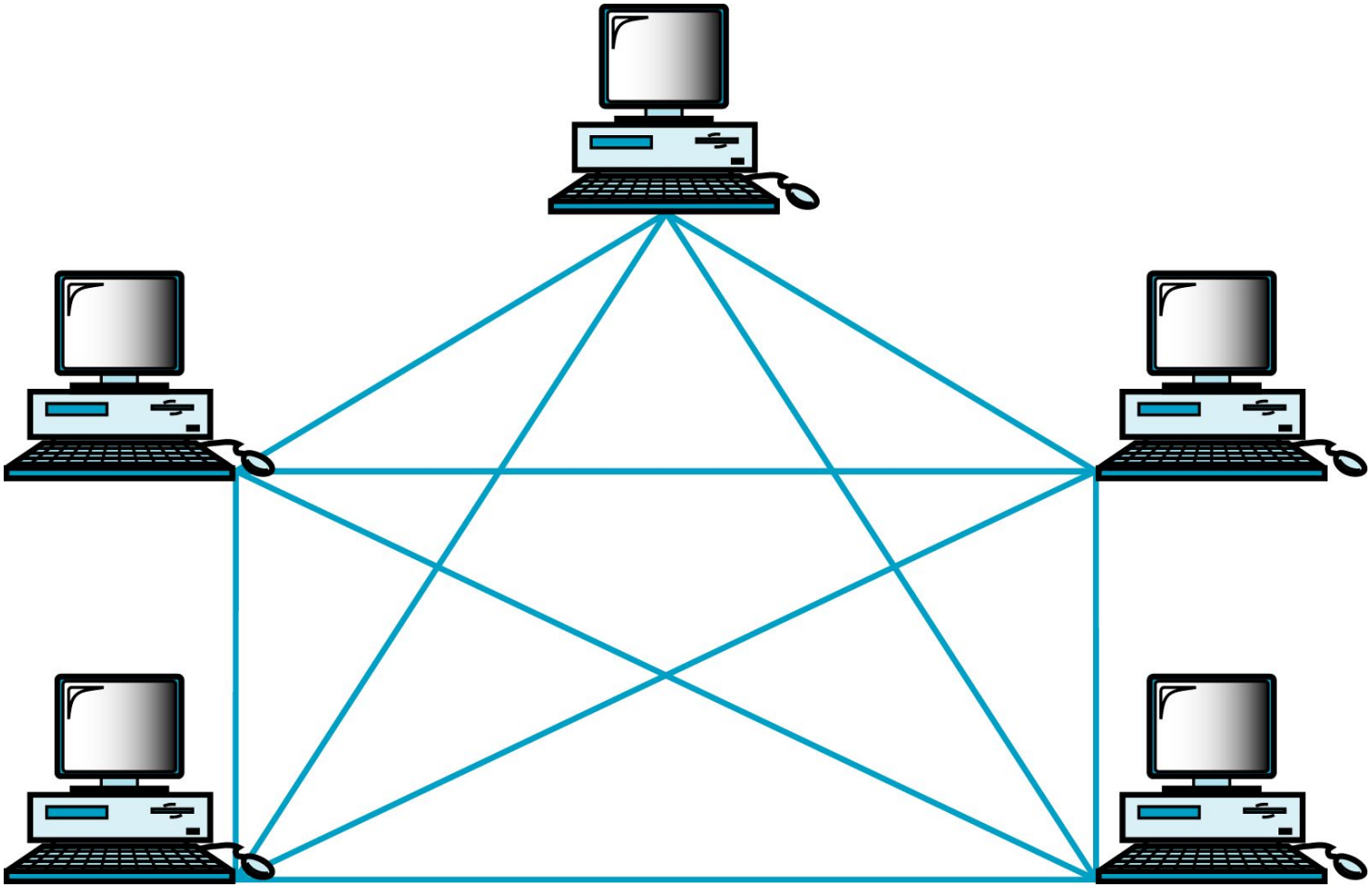
Bus Topology (Cont.)

- **ADVANTAGES**
  - popular because its working requirements are comparatively simple, and growth is accomplished easily and quickly.
  - Easy to use and understand.
  - It is reliable in small networks.
  - Repeater can be used to extend the network.
  - It requires least amount of cable.
  - It is less expensive in cost.

Bus Topology (Cont.)

- **DISADVANTAGES**
  - Heavy network traffic can slow the Bus, considerably (more collision, more waiting time).
  - Each BNC Barrel connecter weakens the signals as it absorbs energy for its own charge.
  - Difficult to troubleshoot and maintain because of cable break, loose connector, and malfunctioning computer.
  - Single computer failure disturbs the whole network.
  - Not a good choice for large organizations, because of slow data traffic.

Mesh Topology



Mesh Topology (Cont.)

- In a mesh topology, every device has a dedicated point-to-point link to every other device.
- The term dedicated means that the link carries traffic only between the two devices it connects.
- Primarily used in networks that are mostly not LANs.
- Made up of multiple point to point connections.
- Mesh networks can get complicated very quickly, because many connections must be managed in a network of any size.
- Mesh topology have redundant links between devices.
- There are two types of Mesh, i.e.
  - ***True Mesh***
  - ***Hybrid Mesh***

Mesh Topology (Cont.)

- **TRUE MESH**
  - A true Mesh has a link between each device in the network.
  - It means that network is fully connected.
  - A true mesh of six devices requires fifteen connections (5 x 6)/2 = 15 links and that for seven devices requires 21 links.
  - The formula for finding total links in a true mesh is ***[n * (n – 1)] / 2,*** where ***n*** is total number of workstations.

Mesh Topology (Cont.)

- **HYBRID MESH**

  - Hybrid Mesh has extra links but not fully connected. Most Mesh topology networks are Hybrid Mesh networks.

Links and Ports in Mesh Topology

- Number of Links required to connect ‘n’ devices: n(n-1)/2
- Number of I/O Ports: n-1
- In figure above, we have 5 Nodes, therefore:
  - No. of Links= 5(5-1)/2 = 10
  - No. of I/O Ports= 5-1 = 4

Mesh Topology (Cont.)

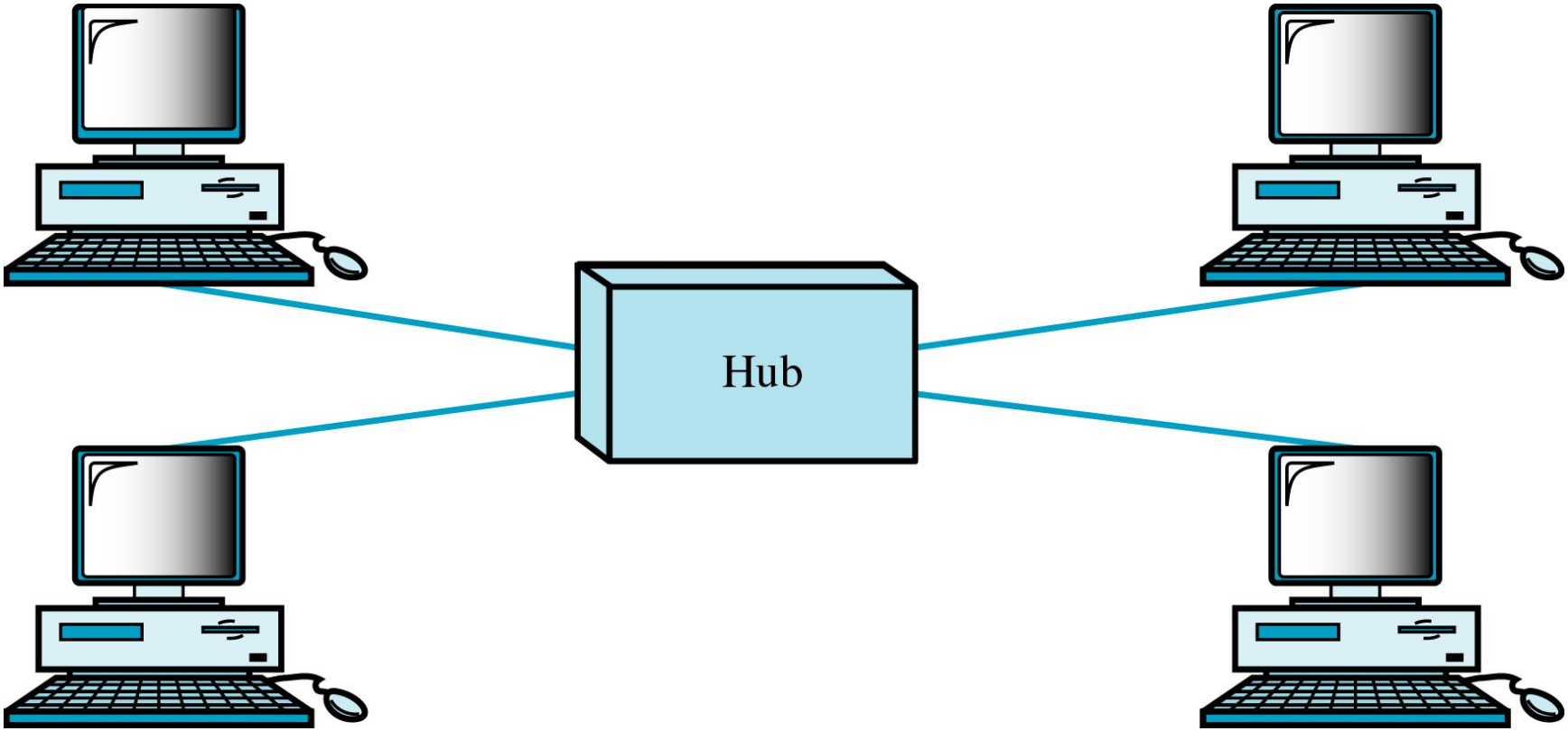
- **ADVANTAGES**
  - Each dedicated connection can carry its own data load, thus eliminating the traffic problems.
  - It is robust. If one link becomes unusable, it does not incapacitate the entire system.
  - privacy or security. Physical boundaries prevent other users from gaining access to messages.
  - Point-to-point links make fault identification and fault isolation easy.

Mesh Topology (Cont.)

- **DISADVANTAGES:**

  - Amount of cabling and the number of I/O ports required
  - Installation and reconnection are difficult.
  - The sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
  - The hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Star Topology



Star Topology (Cont.)

- Most commonly used network topology

- **STRUCTURE:**
  - Each device has a dedicated point-to-point link only to a central controller, usually called a hub or switch.
  - The devices are not directly linked to one another.
  - Unlike a mesh topology, a star topology does not allow direct traffic between devices.
  - The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

Star Topology (Cont.)

- **WORKING:**

  - Each computer on star network communicates with a Hub, that resend the message either to all computers which is called **Broadcasting Star** or only to the destination node which is called **Switched Star**.
  - Now-a-days most of Switched Star networks use Switches instead of Intelligent Hubs because switches are less expensive than Intelligent hubs.

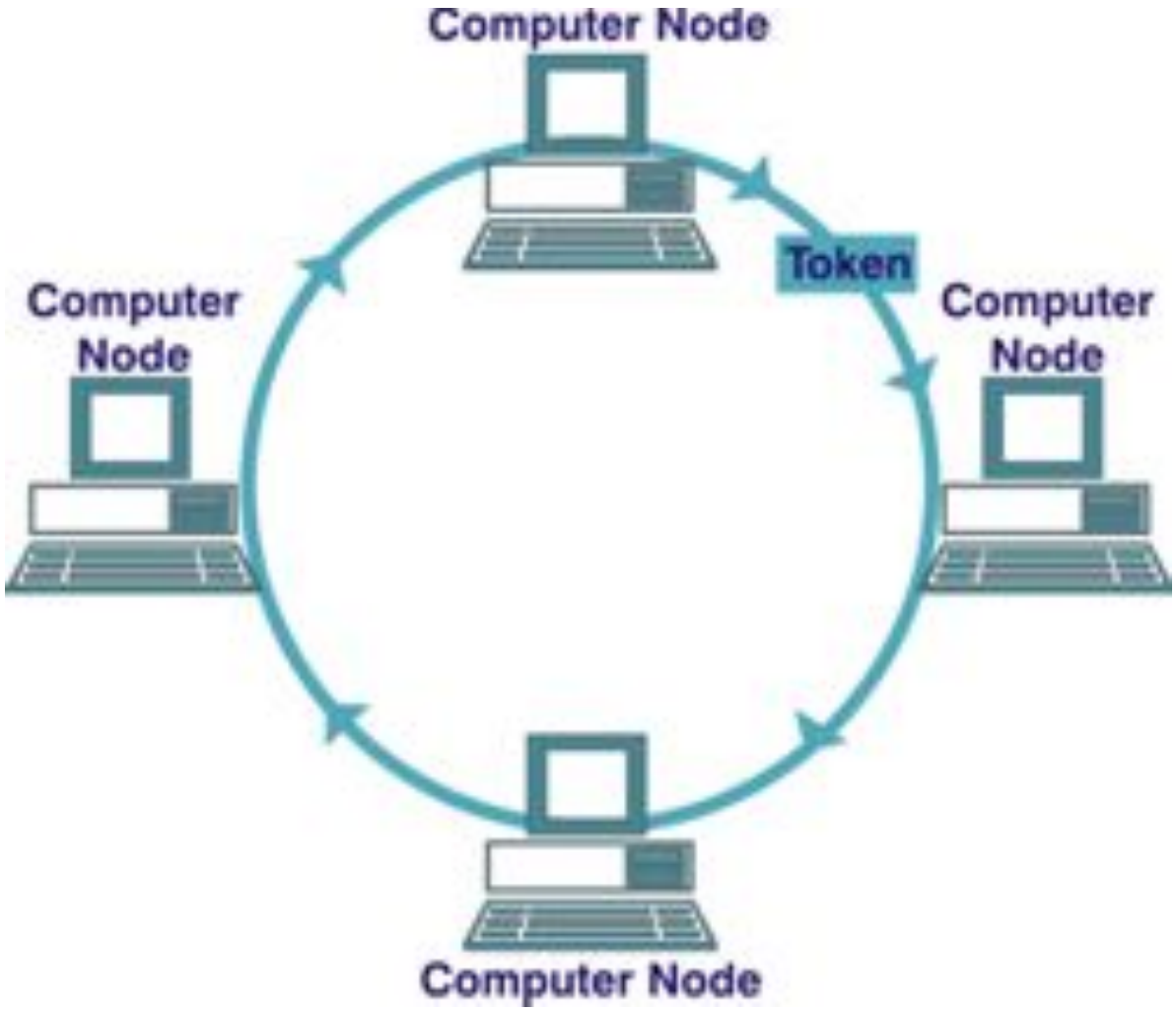
Star Topology (Cont.)

- **ADVANTAGES:**
  - Easy modification and expansion, without disturbing rest of the network.
  - Single computer failure does not affect the whole network.
  - Support of several cables types in the same network.
  - Most flexible of the three topologies and easiest to diagnose.
  - It is less expensive than a mesh topology.
  - Each device needs only one link and one I/O port to connect it to any number of others.
  - This factor also makes it easy to install and reconfigure.
  - Far less cabling needed as compare to mesh topology.
  - Robustness. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to

Star Topology (Cont.)

- **DISADVANTAGES:**
  - Dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.
  - More cables are required as compare to Bus and Ring topology and also Hub is expensive.

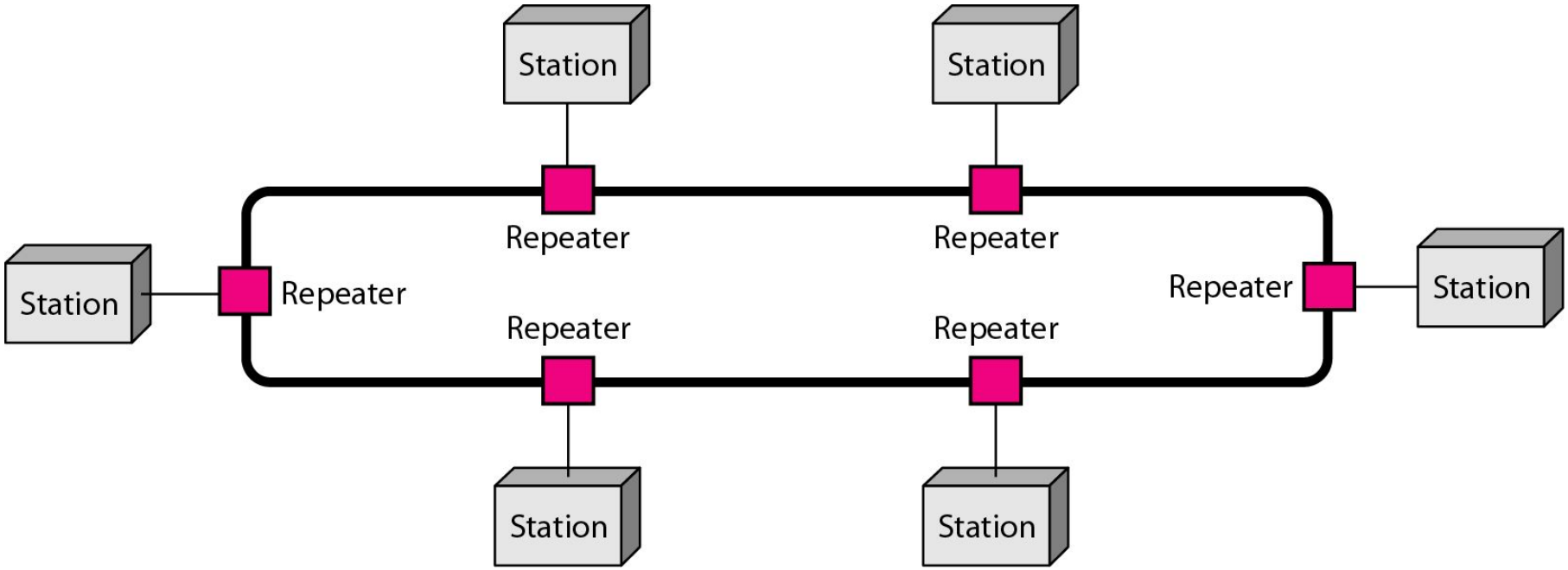
RING TOPOLOGY



RING TOPOLOGY

- Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along
- There is no termination because there is no end to the ring.

RING TOPOLOGY



RING TOPOLOGY

- **Working:**
  - For data transmission, medium access is required that is done by Token and the technique is known as Token Passing.
  - Token is a short message that is passed around the ring until a computer wishes to sends information to another computer, That computer modifies the token, adds an electronic address & data, and sends it around the ring.
  - Each computer in sequence receives the token and the information and passes them to the next computer until either the electronic address matches the address of a computer or the token returns to its origin. The receiving computer returns a message to the originator indicating that the message has been received.
  - The sending computer then creates the token and begins transmitting. The token circulates until a station is ready to send and captures the token.

RING TOPOLOGY

- This all happens very quickly, a token can circle a ring ***200 meters*** in diameter at about ***10,000 times a second***. Some faster networks circulate several tokens at once.
- Transmitted data moves in clockwise direction therefore there is no chance of collision.
- **Example:** ***FDDI*** is a fast fiber-optic network based on the ring topology. Token ring is another ring based network.

RING TOPOLOGY

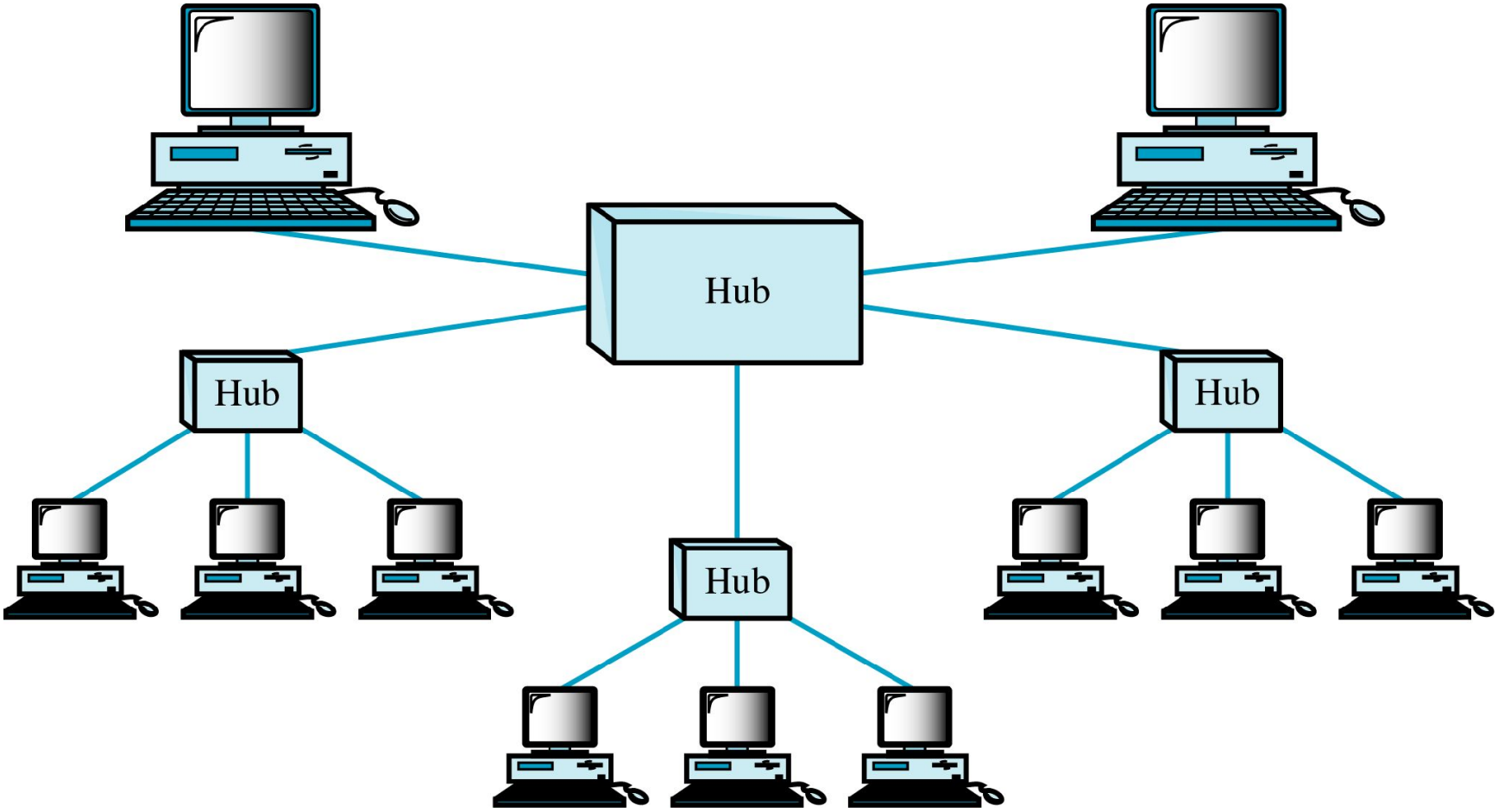
- **ADVANTAGES:**

  - It eliminates LANs depending on a central computer by distributing some of the central systems responsibility to all the other connected workstation.
  - Every workstation on the ring listens to see if the network is clear before sending a message. This helps avoiding collision between the messages.
  - The network degrades gracefully as more users we added i.e. the network doesn’t failed.

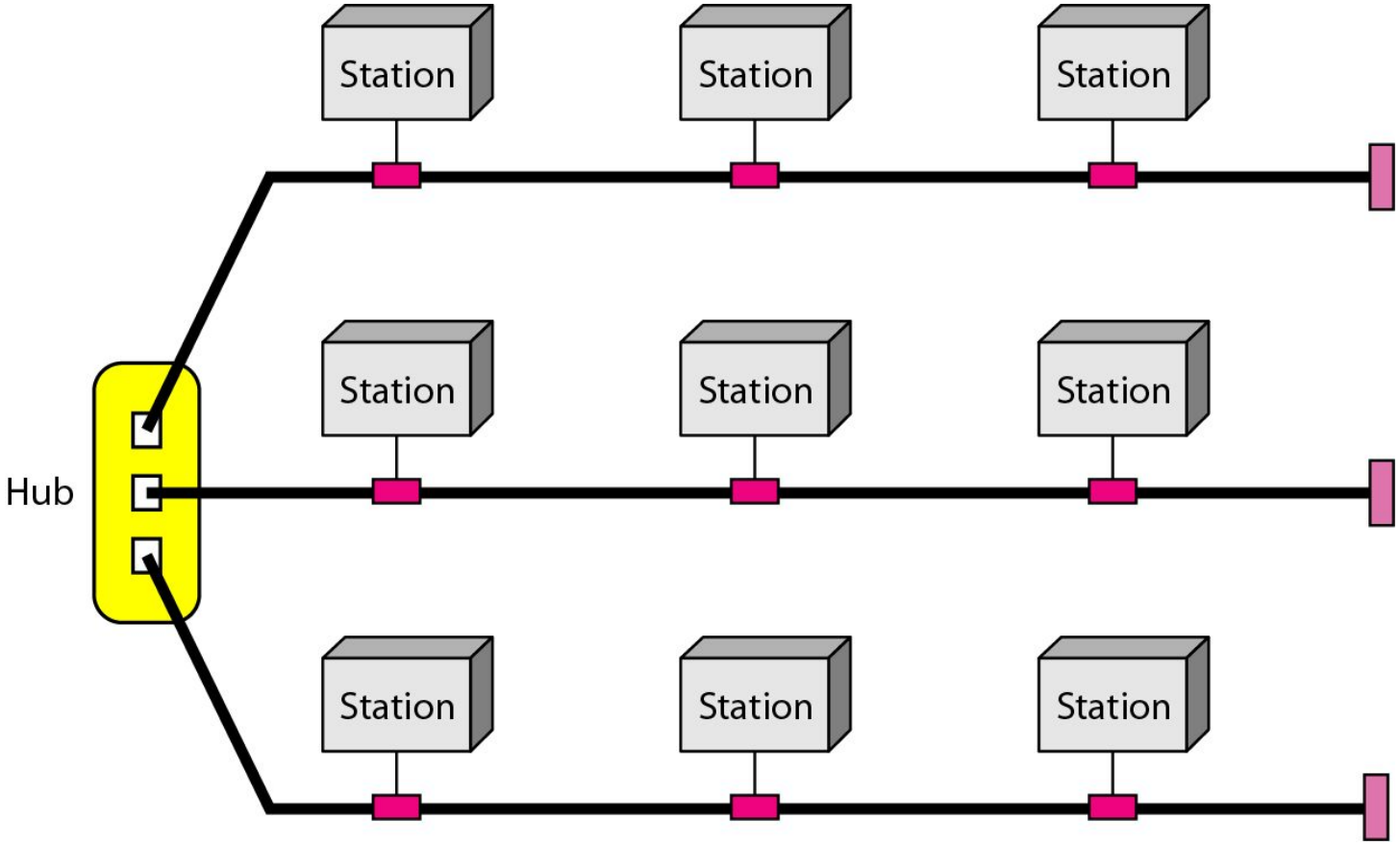
RING TOPOLOGY

- **DISADVANTAGES:**
  - Failure of one computer on the ring can affect the whole network.
  - Also difficult to troubleshoot.
  - Adding or removing nodes disturb the network.

Tree Topology



hybrid topology:
a star backbone with three bus networks



Summary

- Physical Structure of Network
  - Physical Topologies
    1. Bus
    2. Mesh
    3. Star
    4. Ring
    5. Tree
    6. Hybrid

Suggested Reading

- Section
  - 1.2
    - “Data Communications and Networking” 4th Edition by Behrouz A. Forouzan